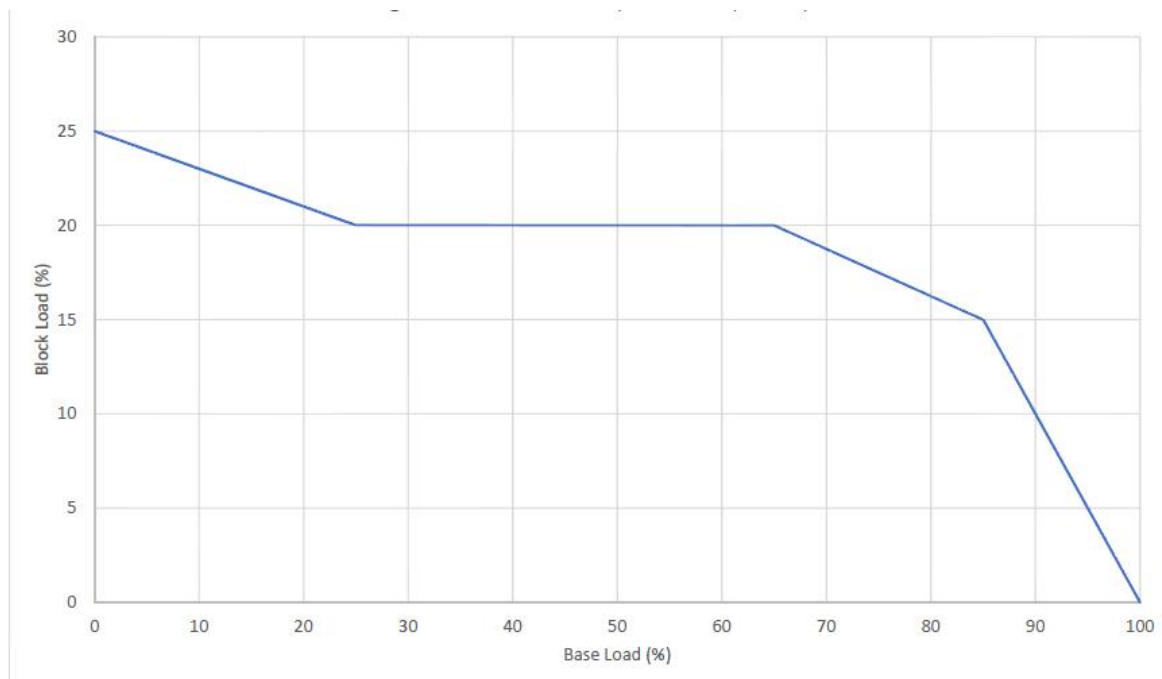


Gas genset technical parameters

Genset Model		KGW550 GF-T
COP@55°C, Working condition	kW	550
Speed	rpm	1000
Exhaust temperature	°C	587
Allowable exhaust back pressure	Mm H ₂ O	368
Fuel consumption(At 100% load)	KJ/kWh	10619
Lubricant consumption	g/kWh	< 0.427
Electrical efficiency(LHV, At 100% load)	%	33.9
Bore/stroke	mm/mm	238/216
Intercooler water inlet/jacket water outlet	°C/°C	60/82.2
Starting mode		Electric starting
PF,COSΦ		0.8 agging
Speed adjustment mode		Electronic governor
Minimum load rate		30%
Generator type		Synchronous
output voltage	V	415
frequency	Hz	50
No. of phase		3 Phases/ 4 Wire
Pilot exciter		PMG
Voltage Variation	%	±10%
Frequency Variation	%	±5%
Transient response characteristic		G4
Noise level		85dB(A) at 1m
Dry weight of genset	kg	19,500
Dimensions with acoustic Enclosure (L*W*H)	mm	12000*3000*3000

Genset Transient Data:

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Note: G4 performance is based on 5 Hz frequency drop for load additions.

Engine Parameter

Item	Unit	Value
Model		F3524 GSIS5
Rated power@55°C, Working condition	kW	590
Type		4-cycle and turbocharging
Cylinders/Arrangement		6/L
Bore	mm	238
Stroke	mm	216
Displacement	L	59
Speed	rpm	1000
Compression ratio		9.7:1
Cooling mode		Closed water cooling
Air/fuel ratio setting		ESM2
Exhaust manifold		Insulated Dry Type
NOx selection (With Emission Control system)	mg/Nm3 ,To 3% O2	72mg/Nm3

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CO (With Emission Control system)	mg/Nm3 ,To 3% O2	<50
Lube oil capacity	L	273
Water capacity	L	214

In Attachment-1 fuel gas source and 55 ° C environment conditions, the engine data sheet is as follows:

Alternator parameter

Power(kW)	550
Exciting mode	Brushless excitation
Phase	three-phases, four-wire
Connection	Star serial
Power factor	0.8
Frequency(Hz)	50
Rated voltage(V)	415
Rated current(A)	957
Rated speed(RPM)	1000
Insulation/ Temperature rise	H/F
Protection grade	IP23

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ALTERNATOR ELECTRICAL DATA LSA 50.2 VL10 / 6P

LS Reference: 550kW 415V 1000rpm

Date: 2024-10-18

V6.10d - 07/2024

Main data:				C		
Power:	688	kVA	550	kWe	579	kWm
Voltage:	415	V	Frequency:	50	Hz	
Rated voltage range:	+5% / -5%		Speed:	1000	rpm	
Power factor - Lagging:	0,8					
Nominal current:	957	A	Phases	3		
Insulation / Temperature rise:	H / F		Connexion	Star serial		
Cooling:	IC01		Winding type:	p2/3		
			Winding:	6S - 6 Wires		
Ambient temperature:	55	°C				
Altitude:	1000	m	Overspeed (rpm)	1200		
Duty: Base Rating			Total Harmonic Distortion (THD) <	3,5%		

Efficiency (Base 550,4 kWe)						IEC
	25%	50%	75%	100%	110%	
Power factor - Lagging: 0,8	91,87	94,69	95,19	95,13	95,04	1
Power factor - Lagging: 1	92,71	95,74	96,44	96,58	96,56	1

Reactances (%) - (Base 688 kVA)							
Unitary impedance (1 per unit) = 0,250509 ohms							
			Unsaturated	Saturated		Unsaturated	Saturated
		Direct axis				Quadrature axis	
Synchronous reactance		Xd	167	138	Xq	85	71
Transient reactance		X'd	17,8	15,1	X'q	85	71
Subtransient reactance		X''d	10,0	8,5	X''q	11,4	9,7
Negative sequence reactance		X2	10,7	9,1			
X0	2,2	Zero sequence reactance					
Xl	5,0	Stator leakage reactance					
Xr	13,9	Rotor leakage reactance					
Kc	0.72	Short-circuit ratio					

Time constants (s)					
		Direct axis		Quadrature axis	
Open circuit transient time constant	T'do	1,16	T'qo	NA	1
Short-circuit transient time constant	T'd	0,124	T'q	NA	1
Open circuit subtransient time constant	T''do	0,011	T''qo	0,041	1
Subtransient time constant	T''d	0,006	T''q	0,005	1
Ta	0,025	Armature winding short circuit time constant			1

Resistances (%)					
Ra	1,4	Armature resistance	R0	0,7	Zero sequence resistance
X/R		X/R ratio (without unit)	R2	2,1	Negative sequence resistance

#REF!

Lube oil System

The engine will be furnished with pressure lubrication system providing oil to all surfaces requiring pressure lubrication. Circulation will be by means of positive displacement engine-driven pump equipped with adjustable pressure regulator. All components will be installed and mounted on the engine or engine base frame complete with piping and including automatic lube oil temperature control. Carbon steel lube oil piping will be pickled. Stainless steel tubing and piping will be cleaned with suitable solvent. Oil cooler will be provided to maintain lube oil supply temperature at or below 65°C. The lube oil pressure at the c

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outlet is greater than the cooling water pressure specified to prevent contamination of the lube oil in case of cooler failure. Oil cooling will be by radiator.

We will be equipped with an automatic oil replenishment system. When the lubricating liquid level in the oil pan is detected to be too low, the oil pump will be transferred to fill the oil pan. The lube oil replenishment pump will pump the lube oil from the lube oil storage tank into the oil pan of the unit, and the capacity of the lube oil storage tank can meet the consumption of 5 days.

Starting System

The starting system includes electric motor, starting controller, starting contactor, a battery charger and a nickel-cadmium battery pack. The battery capacity is 200AH to ensure that the genset is started for a minimum of six consecutive times. The supporting charging system ensures that the generator battery is fully charged and automatically charged after each start operation. The start control of the genset is integrated in the GEGCP. The continuous running time of the electric motor should not exceed 5 seconds each time. If a start fails, it can be restarted after an interval of 15 seconds. No-load operation is strictly prohibited.

Cooling System

The engine is cooled by one closed-circulation horizontal multi-fan water tanks. The water tank is driven by electricity. The radiator area of the tank will be placed on the top of the container. It consists of expansion water tank, cooling water tank, fan, circulating pump, and supporting pipes and valves.

Cooling system will include:

- Remote radiator with expansion tank
- The temperature controller
- Two coolant heating devices (220 V)
- Temperature sensors
- Valves, coolant pipelines
- Coolant drain pipe outside the container
- Manual coolant filling pump with flexible pipelines
- The radiator blowing system will consist of electric fans connected via frequency regulators, the inclusion of which and the rotation speed will be adjusted automatically, depending on the temperature of the coolant and the environment.
- There will be a manual mode for turning on the fans
- Fans will be covered with protective barriers

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- Provide visual monitoring of the coolant level

Inlet and Exhaust System

We will supply industrial type air intake filters with replaceable dry elements to be mounted inside the acoustic enclosure. Each engine will have one air intake filters.

Each engine will have its own exhaust system. The exhaust system includes silencer, exhaust pipe, wave expansion joint, silencer bracket, etc. The silencer is an integrated fire-fighting flower-shaped structure and be completed with companion flanges and rain cap. After the exhaust noise is attenuated by the silencer, the noise is $\leq 85\text{dB (A)}$ at a distance of 1m from the exit.

Fuel System

The fuel intake system is a skid-mounted complete set, including a dual-type main filter, a primary shut-off valve, a secondary shut-off valve, a check valve, two stage pressure reducing valve group etc. All filters are equipped with differential pressure indicators. The primary shut-off valve adopts an electric two-way valve, which is connected to the automatic exhaust valve downstream of the valve, and the valve is operated by the genset control system to start and close the logic operation.

Fuel Consumption Table

Description	Unit	Value
Intake pressure	bar	7
Allowable volatility	$\pm\%$	(of the set value) 10
Maximum Volatility	kPa/sec	1
Consumption	Nm ³ /h	160

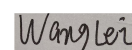
Instrumentation and Controls

Engine controls will be provided to protect the engine against the followings:

- (a) Low lube oil pressure;
- (b) High temperature cooling water leaving the engine jackets;
- (c) Over-speed;
- (d) Excessive vibration;
- (e) High lube oil temperature.

Each unit is equipped with a monitoring cabinet, which is installed in the monitoring room separated by the computer room, which will display the following parameters:

- (a) Mechanical Tachometer;
- (b) Oil Pressure Gauge;
- (c) Water Temperature Gauge (water out of engine);



- (d) Engine Manifold Vacuum Gauge;
- (e) Running Time Meter, (5 digits minimum);
- (f) Start/Stop Control;
- (g) Exhaust Gas Temperature;
- (h) Cooling Water Pressure.

Acoustic Enclosure

The acoustic Enclosure is designed according to ISO 15664 noise level requirements. I protection will be a minimum of IP54. The maximum noise level not exceed 85 dB (A) at 1m from the enclosure.

LCP

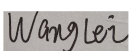
- a) Control cabinet size: 800 * 600 * 2260 (mm) (D * W * H)
- b) Control cabinet color: RAL7035
- c) Control cabinet protection grade: IP42
- d) Protection, control and metering equipment
- e) Complete AC sub-distribution for the generator space heater, panel space heater and any other generator auxiliary items
- f) Raise/lower voltage adjuster (behind the panel door)
- g) Generator status, alarm and trip functions
- h) Automatic starting / synchronizing facilities with manual override, synchroscope and synchronizing lamps
- i) Control of switchgear remote generator CB
- j) Control gear for genset auxiliaries
- k) Control relays (under-voltage, etc...)
- l) Battery backed up DC power supply distribution for the generator protection systems (may be combined with the engine protection systems)
- m) Generator testing facility
- n) Winding/bearing temperature monitoring facility Vendor shall provide Redundant serial communication (RS485) for interface from Gas Generator Vendor provided LCP to Control system (UCP).
- o) Fire Alarm & Detection and Fire Fighting System Equipment, Panel and Interface.

GCP

Placed inside the Substation. Includes control and monitoring equipment as well as engine measurement/control, all components within the gas generator housing as well as exte

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fans are controlled via a panel-mounted HMI. We will provide a redundant serial communication (RS485) interface from the gas generator LCP to the control system (UCP).

A handwritten signature in black ink, reading "Wang Lei".